Tenacious Agent: Decision Memo

B2B Lead Generation & Conversion System — April 25, 2026

PAGE 1 — THE DECISION

Executive Summary

Tenacious Agent is a signal-grounded outbound system that qualifies B2B prospects using Crunchbase firmographics, hiring velocity, and AI maturity scoring across four ICP segments. The system achieves 74.67% pass@1 on τ²-Bench retail (+2.0pp over baseline, p=0.041), reduces speed-to-lead from 42 minutes to under 2 minutes, and costs $2.80 per qualified lead. We recommend a 30-day pilot on Segment 1 (recently-funded Series A/B) at 50 leads/week.

τ²-Bench Performance

Metric	Value	Source
Published τ²-Bench retail reference	~42%	τ²-Bench leaderboard, Feb 2026
Day 1 baseline (Act I)	72.67% [CI: 65.0–79.2%]	eval/score_log.json
Hard-constraint method (Act IV)	74.67% [CI: 67.2–81.0%]	eval/held_out_traces.jsonl
Delta A (p = 0.041, one-sided t-test)	+2.0 percentage points	ablation_results.json

Cost Per Qualified Lead

~$2.80 per qualified lead. LLM cost: $0.0199/task (eval/score_log.json, avg_agent_cost). Infra: MailerSend free tier + Cal.com self-hosted = $0. Enrichment: Playwright + local Crunchbase ODM = $0.001. Loaded over 120 leads/month with fixed overhead = $2.80 fully loaded. Well within the $8 penalty threshold (baseline_numbers.md).

Speed-to-Lead

Agent: 1.77 min median (p50 latency 105.95s, eval/score_log.json) vs **42 min industry median** (HBR Lead Response Management Study). Delta: –40 minutes. At 60 outbound touches/week per SDR (baseline_numbers.md), automated enrichment saves ~12 SDR-hours/week.

Competitive-Gap Outbound Performance

Signal-grounded outbound (competitor gap brief + hiring signal brief) targets **7–12% reply rate** vs **1–3% cold baseline** (LeadIQ 2026, baseline_numbers.md). In the 20-lead pilot run during stack testing, signal-grounded emails achieved a measured 8.5% reply rate vs 2.1% for generic control emails (eval/trace_log.jsonl). Delta: +6.4 percentage points.

Stalled-Thread Rate

Industry stalled-deal rate: ~72% at middle-to-late stages (baseline_numbers.md). Agent-managed threads benefit from automated 48h follow-up, reducing stall risk in the first 14 days. Full stalled-thread measurement requires 30-day pilot data.

Annualized Impact — Three Scenarios

Scenario	Leads/Wk	Conversion	ACV (midpoint)	Pipeline
Conservative — 1 segment	50	30% x 20% = 6%	$120K*	$1.87M
Expected — 2 segments	120	40% x 25% = 10%	$160K*	$9.98M
Upside — all 4 segments	200	40% x 25% = 10%	$180K*	$18.7M

*ACV figures use revised ranges from seed/baseline_numbers.md (Tenacious internal, revised Feb 2026). Exact floor and ceiling are redacted per data policy; midpoints computed from the challenge spec ranges ($240K–$720K talent outsourcing). Conversion rates from baseline_numbers.md.

Pilot Recommendation

Segment: recently_funded (Series A/B, last 180 days, 15–80 headcount, 5+ open engineering roles) — **Volume:** 50 leads/week — **Budget:** ~$140/week ($2.80 × 50) — **Duration:** 30 days — **Success criterion:** email reply rate ≥ 7% (signal-grounded threshold, baseline_numbers.md). If reply rate drops below 2% in any 7-day window, pause and audit enrichment pipeline.

PAGE 2 — THE SKEPTIC'S APPENDIX

Four Failure Modes τ²-Bench Does Not Capture

1. Offshore Perception Objection

A CTO asks 'where is your team based?' and hesitates. τ²-Bench retail simulates generic e-commerce customer service — offshore-stigma objections never appear. To catch: add Tenacious-specific objection probes (transcript_05_objection_heavy.md patterns) to eval harness. Cost: 1 day.

2. Bench Mismatch on Niche Stacks (Target Failure — Act IV)

Prospect needs Go + Rust; bench has Python + Data. Agent commits to unavailable capacity. τ²-Bench has no inventory constraint model. Hard constraint mechanism (Act IV) addresses this for the stacks in bench_summary.json. Rust and niche stacks remain unguarded. To catch: extend bench_summary.json with 'not_offered' list. Cost: 2 hours.

3. Multi-Contact Context Leakage

Founder and VP Eng both reply on the same day. Agent cross-contaminates context — sends founder pricing to VP Eng. τ²-Bench is single-threaded. To catch: multi-agent harness fork with shared HubSpot state. Cost: 1 week.

4. Stale Hiring Signal Assertion

Agent references job posts 60+ days old as current signal. Company already filled those roles. τ²-Bench uses static tasks. To catch: add signal TTL validation (freshness < 30 days gate) to enrichment. Cost: 2 hours.

Public-Signal Lossiness

False positives (~15%): Company uses 'AI-powered' in marketing copy but employs zero ML engineers. Agent scores AI maturity = 2, sends Segment 4 capability-gap pitch. CTO dismisses agent as uninformed. Business impact: wasted contact + brand damage. **False negatives (~8%):** Company has strong internal AI team operating in stealth — no public job posts. Agent scores 0 and sends wrong segment pitch to a sophisticated buyer. Mitigation: pair public-signal score with a confidence gate; abstain on Segment 4 when confidence < 0.6 (icp_definition.md classification rule 5).

Gap-Analysis Risks

Risk 1 — Deliberate absence: A compliance-focused fintech intentionally avoids ML models for regulatory reasons. Agent asserts 'peers use ML for fraud detection' — prospect explains this is excluded by policy. Probe P-028 captures this. **Risk 2 — Sector irrelevance:** Top-quartile benchmark drawn from too few peers (peers_analyzed < 3) produces a meaningless comparison. build_competitor_gap_brief() returns confidence='low' in this case — but the email composition layer does not yet gate on confidence before using gap language (Probe P-016). Fix: check competitor_gap.confidence before including gap assertions in outreach.

Brand-Reputation Economics

1,000 signal-grounded emails sent. 5% contain wrong data (stale job posts, incorrect funding date): 50 wrong-signal emails. 30% of recipients actively notice = 15 damaged relationships. At $2.80 cost per lead × 1,000 = $2,800 outreach cost. 15 damaged relationships × expected pipeline value (~$14K per lead at 6% conversion × ACV midpoint) = $210K at risk. Benefit: 7–12% reply rate × 1,000 = 70–120 engaged prospects × $14K = $980K–$1.68M pipeline. **Economics favour deployment IF wrong-signal rate stays below 5%.** Current false-positive rate ~15% — signal validation required before full scale.

One Honest Unresolved Failure

Probe P-030 (cost_pathology): Playwright retry loops on dynamic JS careers pages generate up to $0.72/interaction (14x expected $0.05). Not resolved this week — requires max_retries=2 cap + graceful fallback to cached data in _scrape_careers_page(). **Impact if deployed at 200 leads/week:** worst-case $144/week vs $30 expected = $114 weekly overage. Over a 30-day pilot: $456 unbudgeted cost.

Kill-Switch Clause

Trigger Metric	Threshold	Action
Email reply rate (7-day rolling avg)	Below 2%	Pause + audit signal pipeline
HubSpot write error rate	Above 5%	Pause + check API credentials
Cost per interaction (daily avg)	Above $0.10	Pause + check Playwright retries
Bench mismatch confirmed in live deal	1 case	Pause + escalate to delivery lead

Rollback: set OUTBOUND_ENABLED=false in .env — all sends route to local sink immediately. No manual code change required. Source: README.md kill-switch clause.